NATIONAL INSTITUTE OF TECHNOLOGY PATNA
MID-SEMESTER EXAMINATION, SEPTEMBER-2025

Program: B.Tech-CSE-DD(DS-CS)
Department: CSE
Full Marks: 30
Duration of Examination: 2 Hours

Semester: 5th
Course Code: CS54116
Course Name: Software Engineering

INSTRUCTIONS

- *Please mention your section at the top of the answer script.*
- *Answer all the questions. Assume Missing data, if any.*
- *Each question includes the corresponding marks, Course Outcome (CO), and Bloom's Level (BL).*

1. Draw a schematic diagram to represent the iterative waterfall model of software development. On your diagram represent the following:
   I. The phase entry and exit criteria for each phase.
   II. The deliverables that need to be produced at the end of each phase. *(5M)[CO1][L2]*

2. Suppose you are the project manager of a software project (Named As: Shop Automation S/W) requiring the following activities.

Activity No.	*Activity Name*	*Effort in person-months*	*Immediate Predecessor*
1	Obtain requirements	3	-
2	Analyse operations	3	-
3	Define subsystems	1	1
4	Develop database	2	1
5	Make decision analysis	3	2
6	Identify constraints	1	5
7	Build module 1	5	3,4,6
8	Build module 2	8	3,4,6
9	Build module 3	12	3,4,6
10	Write report	10	6
11	Integration and test	8	7,8,9
12	Implementation	2	10,11

   (a) Draw the Activity Network representation of the project.

   (b) Determine ES, EF and LS, LF for every task.

   (c) Draw the Gantt chart representation of the project. *(2+6+2M)[CO2][L3,L4]*

3. A metropolitan city plans to deploy a Traffic Management System (TMS) to reduce congestion and improve safety. The system should collect live data from traffic cameras, sensors, and GPS devices in vehicles. It should process this data to monitor traffic density, detect accidents, and adjust traffic signals dynamically. Drivers should be able to view real-time traffic updates, alternate routes, and estimated travel times via a mobile app. Law enforcement officers should receive automatic alerts in case of accidents, violations, or emergencies. The system should also generate daily traffic reports for city administrators. The system must handle real-time data streams from millions of devices. Processing and response time for critical alerts must be less than 1 second. It should have 99.95% uptime and disaster recovery mechanisms. The system should ensure data privacy regulations to protect citizen information. The mobile app should be user-friendly, multilingual, and accessible for differently abled users. The system should be scalable to support future features like smart parking,

pollution monitoring, and integration with autonomous vehicles. Now extract and elaborate all the functional, nonfunctional and other non-functional requirements of the TMS. *(4+4+2M)[CO2][L4]*

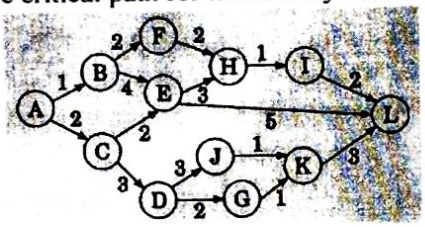
4. Mark the correct option (s) *(1+2+1+1M)[CO2][L3,L4]*

I. Using the data of the following table, what can be said about the total project duration?

Activity	Duration	Predecessors
A	14	-
B	13	A
C	20	A
D	10	B, C

(a) 20
(b) 27
(c) 37
(d) 44

II. Which one of the following is the critical path for the activity network given below?



(a) A, B, F, H, G, I, L
(b) A, B, F, H, I, L
(c) A, C, D, J, K, L
(d) A, C, D, G, K, L

III. An activity has an early start (ES) of day 3, a late start (LS) of day 13, an early fi nish (EF) of day 16, and a late fi nish (LF) of day 26. Which one of the following can be inferred for this activity?
(a) It is on the critical path
(b) It is not progressing well
(c) It is progressing well
(d) It is not on the critical path

IV. Which one of the following is true for prototypes constructed in the prototyping model of software development?

(a) Construction of prototypes incurs significant cost and should be avoided as far as possible.
(b) A prototype is useful for eliciting early feedback on requirements for the stakeholders.
(c) Prototypes are usually constructed toward the end of the design phase.
(d) Prototypes increase the risk of misunderstandings between developers and users.